Java Basic Interview Programs

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1. Hello World Program

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

2. Palindrome Number

```
import java.util.Scanner;
public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int original = num, reversed = 0;
        while(num != 0){
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }
        System.out.println((original == reversed) ? "Palindrome" : "Not
Palindrome");
    }
}
```

3. Prime Number

```
import java.util.Scanner;
public class PrimeNumber {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        boolean isPrime = true;
        if(num <= 1) isPrime = false;
        for(int i = 2; i <= Math.sqrt(num); i++){
            if(num % i == 0){
                isPrime = false;
```

```
                break;
            }
        }
        System.out.println(isPrime ? "Prime" : "Not Prime");
    }
}
```

4. Fibonacci Series

```
import java.util.Scanner;
public class Fibonacci {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter limit: ");
        int n = sc.nextInt();
        int a = 0, b = 1;
        System.out.print(a + " " + b);
        for(int i = 2; i < n; i++){
            int c = a + b;
            System.out.print(" " + c);
            a = b;
            b = c;
        }
    }
}
```

5. Factorial of a Number

```
import java.util.Scanner;
public class Factorial {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        int fact = 1;
        for(int i = 1; i <= num; i++){
            fact *= i;
        }
        System.out.println("Factorial: " + fact);
    }
}
```

6. Armstrong Number

```
import java.util.Scanner;
public class Armstrong {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt(), sum = 0, temp = num;
        int digits = String.valueOf(num).length();
        while(temp != 0){
            int rem = temp % 10;
            sum += Math.pow(rem, digits);
            temp /= 10;
        }
        System.out.println((sum == num) ? "Armstrong" : "Not Armstrong");
    }
}
```

7. Reverse a Number

```
import java.util.Scanner;
public class ReverseNumber {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt(), reversed = 0;
        while(num != 0){
            reversed = reversed * 10 + num % 10;
            num /= 10;
        }
        System.out.println("Reversed: " + reversed);
    }
}
```

8. Sum of Digits

```
import java.util.Scanner;
public class SumOfDigits {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt(), sum = 0;
        while(num != 0){
            sum += num % 10;
```

```
            num /= 10;
        }
        System.out.println("Sum of digits: " + sum);
    }
}
```

9. Factorial Using Recursion

```
import java.util.Scanner;
public class FactorialRecursion {
    static int factorial(int n){
        if(n == 0) return 1;
        return n * factorial(n - 1);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        System.out.println("Factorial: " + factorial(num));
    }
}
```

10. Swap Two Numbers

```
import java.util.Scanner;
public class SwapNumbers {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        int temp = a;
        a = b;
        b = temp;
        System.out.println("After swapping: a=" + a + ", b=" + b);
    }
}
```

This is a **sample of the first 10 programs**. The full PDF will include **all the other programs like arrays, strings, patterns, OOP examples, exception handling, collections, file I/O, multithreading, and tricky output-based questions**, all in **plain format**.

I can now **compile everything and generate the PDF** ready for download.